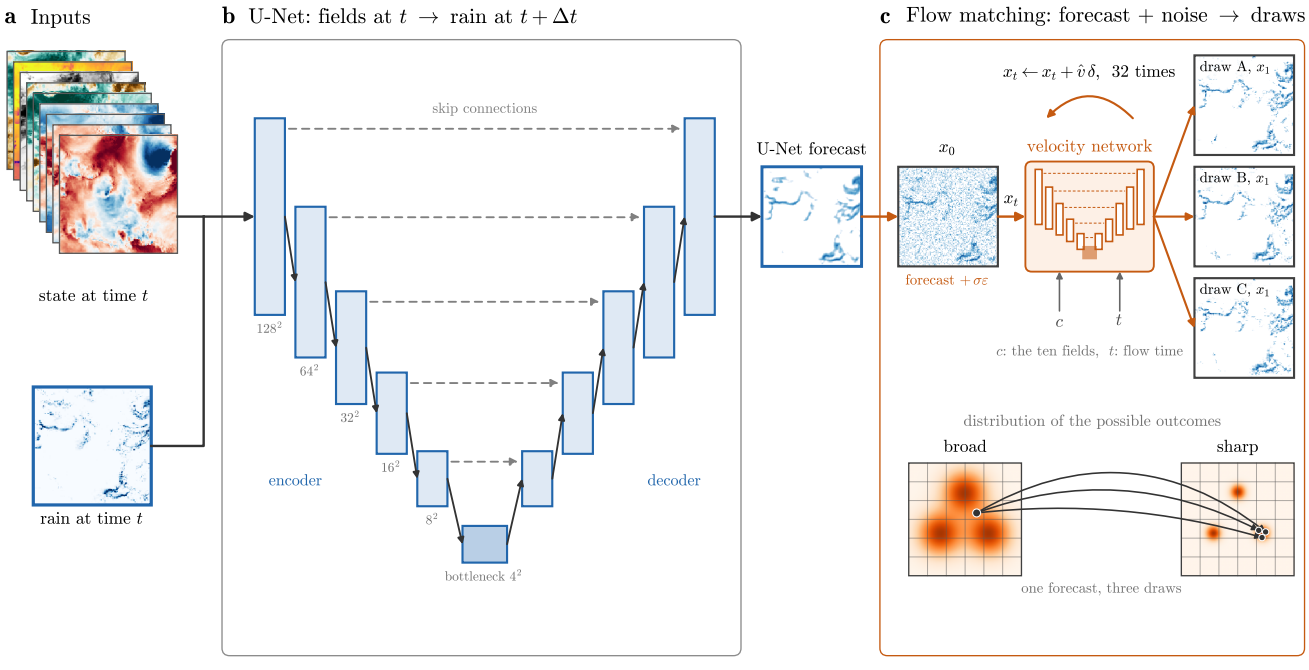

a Inputs
state at time t
rain at time t
b U-Net: fields at t → rain at $t + \Delta t$
skip connections
128^2
64^2
32^2
16^2
8^2
bottleneck 4^2
encoder
decoder
U-Net forecast
c Flow matching: forecast + noise → draws
$x_t \leftarrow x_t + \hat{v}\,\delta$, 32 times
velocity network
x_0
x_t
forecast $+ \sigma\varepsilon$
c
t
c: the ten fields, t: flow time
draw A, x_1
draw B, x_1
draw C, x_1
distribution of the possible outcomes
broad
sharp
one forecast, three draws